

SC Honors Geometry - Nappier
Homework 11.1 – Circumference and Arc Length

Name: _____

Circumference. Give exact answers AND the nearest tenth.

1. A circle has radius 11. Find its circumference.
2. A circle has diameter 26. Find its circumference.

Arc length.

3. Find the length of a 90° arc in a circle of radius 10.
4. Find the length of a 144° arc in a circle of radius 5.
5. Find the length of a 45° arc in a circle of radius 16.
6. Find the length of a 150° arc in a circle of radius 14.

Working backward.

7. An arc of length 6π sits in a circle of radius 9. Find the central angle.
8. A circle has circumference 36π . Find its diameter.

Radians and rolling.

9. Convert 225° to radians (exact), and convert $3\pi/2$ radians to degrees.
10. A wheel has diameter 26 inches. About how many revolutions does it make in one mile (63,360 in.)?